

KOFI OFORI

Research Assistant – Engineering

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Kumasi, Ghana

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PROFESSIONAL SUMMARY

A motivated and detail-oriented Engineering graduate with strong foundations in mechanical and structural engineering, data analysis, and laboratory research. Eager to contribute to the research agenda of Pentecost University's Engineering faculty by supporting experimental design, data collection, and analysis of engineering systems. Committed to academic rigour, collaborative inquiry, and the application of engineering principles to real-world challenges in Ghana and beyond.

EDUCATION

Master of Science – Mechanical Engineering

2020 – 2022

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

- Thesis: "Thermal Performance Analysis of Locally Fabricated Solar Water Heaters in the Ashanti Region"
- Graduated with Distinction; Department of Mechanical Engineering Award for Academic Excellence
- Coursework included: Advanced Thermodynamics, Finite Element Analysis, Research Methodology, Engineering Simulation

Bachelor of Science – Mechanical Engineering

2015 – 2019

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

- First Class Honours; Best Graduating Student in the Department of Mechanical Engineering
- Final Year Project: "Design and Fabrication of a Low-Cost Cassava Processing Machine for Small-Scale Farmers"
- Active member of the KNUST Engineering Student Society (KESS)

RESEARCH & WORK EXPERIENCE

Junior Research Officer – Engineering

Jan 2023 – Present

Council for Scientific and Industrial Research – BRRI (Building and Road Research Institute), Kumasi

- Conduct laboratory and field tests on construction materials including concrete, asphalt, and laterite under supervision of senior researchers
- Assist in data collection and statistical analysis for ongoing infrastructure durability studies funded by the Ministry of Roads and Highways
- Contribute to preparation of technical reports, research papers, and project documentation for institutional and external stakeholders
- Operate and maintain testing equipment including universal testing machines, compressive strength testers, and moisture analysers
- Co-authored two internal technical reports on pavement failure in urban road networks in the Ashanti Region

Research Assistant (Postgraduate)

2020 – 2022

Department of Mechanical Engineering, KNUST

- Assisted principal investigator in designing and conducting solar energy efficiency experiments across 12 field sites
- Collected, cleaned, and analysed thermal performance datasets using MATLAB and Excel
- Contributed literature review sections and methodology write-ups to two submitted journal articles
- Delivered a seminar presentation on low-cost renewable energy technologies at the Faculty Research Day 2021

Engineering Intern

Jun – Sep 2018

- Supported engineering team in conducting site surveys and road condition assessments on selected national highways
- Assisted with AutoCAD drafting of road alignment and drainage designs
- Prepared weekly field reports summarising findings and anomalies for senior engineers

PUBLICATIONS & PRESENTATIONS

- Ofori, K., & Asiedu, B. (2022). Thermal efficiency of locally fabricated flat-plate solar water heaters: A case study in the Ashanti Region. *Journal of Renewable Energy Research Ghana*, 5(1), 23–38.
- Ofori, K. (2021). Low-cost solar technologies for rural households in Ghana. Oral presentation, KNUST Faculty of Engineering Research Day, Kumasi.
- Ofori, K., Boateng, J., & Mensah, R. (2023). Assessment of premature pavement failure on urban roads: Causes and remediation strategies. *BRRI Internal Technical Report No. TR-2023-07*.

TECHNICAL SKILLS

Engineering Software	AutoCAD, MATLAB, ANSYS (FEA), SolidWorks, MS Project
Data Analysis	Excel (Advanced), SPSS, Python (NumPy, Pandas, Matplotlib)
Lab Competencies	Material testing, soil analysis, compressive/tensile strength measurement, environmental sampling
Research Methods	Experimental design, quantitative analysis, technical writing, literature review
Languages	English (Fluent), Twi (Native), French (Basic)

CERTIFICATIONS & TRAINING

- MATLAB Fundamentals Certification – MathWorks Online Academy, 2022
- Research Ethics and Integrity Training – KNUST Office of Research, Innovation and Development, 2021
- AutoCAD 2D & 3D Professional Certificate – Aptech Computer Education, Kumasi, 2019
- Workplace Safety and Health for Engineers – Ghana Institution of Engineering (GhIE), 2023

PROFESSIONAL AFFILIATIONS & SERVICE

- Graduate Member, Ghana Institution of Engineering (GhIE)
- Student Member, American Society of Mechanical Engineers (ASME) – International
- Volunteer Engineering Tutor, SOS Technical Skills Programme, Kumasi (2022 – Present)
- Peer Reviewer, KNUST Journal of Engineering and Applied Sciences (2023 – Present)

REFEREES

Prof. Emmanuel Asiedu Associate Professor & Head of Department, Mechanical Engineering, KNUST e.asiedu@knust.edu.gh +233 32 206 0000	Dr. Joyce Boateng Senior Research Scientist, Building & Road Research Institute (BRRI), CSIR j.boateng@brri.csir.org +233 32 203 1111
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